

Practice Questions Solutions 1

This sheet provides practice with simple linear regression. You will work with the regression model algebraically, fit and interpret a model in **R**, and reproduce key regression calculations by hand. These practice questions should be attempted in your own time. These questions are **not graded** and **should not be handed in**

You should attend the joint ST302/ST303 lab to get help with the questions and have your answers checked.

Student name: _____

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Simple linear regression

Consider the following simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n.$$

The above model may be alternatively written as

$$y_i = \alpha_0 + \alpha_1(x_i - \bar{x}) + \epsilon_i, \quad i = 1, \dots, n.$$

Answer the following:

1. Using the method of least squares, derive expressions for the estimators $\hat{\alpha}_0$ and $\hat{\alpha}_1$
2. Interpret the parameters α_0 and α_1 .
3. Explain why mean-centering the predictor variable can be useful.

Solutions

Question 1

We minimise the residual sum of squares

$$SS_E = \sum_{i=1}^n (y_i - \hat{\alpha}_0 - \hat{\alpha}_1(x_i - \bar{x}))^2$$

Taking the derivative with respect to $\hat{\alpha}_0$ gives us

$$\frac{\partial SS_E}{\partial \hat{\alpha}_0} = -2 \sum_{i=1}^n (y_i - \hat{\alpha}_0 - \hat{\alpha}_1(x_i - \bar{x}))$$

Setting this equal to 0

$$\sum_{i=1}^n y_i - n\hat{\alpha}_0 - \hat{\alpha}_1 \sum_{i=1}^n (x_i - \bar{x})$$

Since

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

We are left with

$$\hat{\alpha}_0 = \bar{y}.$$

Now we can take the derivative with respect to $\hat{\alpha}_1$:

$$\frac{\partial SS_E}{\partial \hat{\alpha}_1} = -2 \sum_{i=1}^n (y_i - \hat{\alpha}_0 - \hat{\alpha}_1(x_i - \bar{x}))(x_i - \bar{x})$$

Setting this equal to 0 leaves us with

$$0 = \sum_{i=1}^n y_i(x_i - \bar{x}) - \hat{\alpha}_0 \sum_{i=1}^n (x_i - \bar{x}) - \hat{\alpha}_1 \sum_{i=1}^n (x_i - \bar{x})^2$$

Again, because

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

We can say that

$$\hat{\alpha}_1 = \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$$\text{Hence: } \hat{\alpha}_0 = \bar{y} \text{ and } \hat{\alpha}_1 = \frac{S_{xy}}{S_{xx}}$$

Question 2

- α_0 is the mean of Y when $X = \bar{x}$.
- α_1 is the slope or change in mean of Y for a unit increase in X .

Question 3

Mean-centering X can make the intercept easier to interpret.

In the original model, β_0 represents the mean response when $X = 0$. If $X = 0$ is outside the observed range, this may not be a meaningful quantity.

After centering: $X - \bar{x} = 0$ when $X = \bar{x}$

Therefore, α_0 represents the mean response at the average value of X .

Simple linear regression in R

As concrete cures, it gains strength overtime. The following data (see Table 1) represent the 7-day and 28-day strength, in pounds per square inch (psi), of a certain type of concrete.

Table 1: Concrete data

7 Day	28 Day
2300	4370
2430	4640
2890	4620
3120	4900
3380	5020
3390	5220

You can read the data from the RStudio server using:

```
# Using the `readr` and the `here` package.  
# TIP: you can press tab when writing the folder path to have all the options  
# folder / file options appear to you  
  
concrete <- readr::read_csv(here::here("Sharedfiles/ST303/data/Concrete.csv"))
```

Answer the following set of questions using R

1. Create a scatterplot showing the relationship between 7-day strength on the x -axis and 28-day strength on the y -axis.
2. Does it seem appropriate to assume a linear relationship between the two variables (explain why you think so)?
3. Specify a simple linear regression model for these data algebraically.
4. Fit this model in R and report what the **intercept**, **slope**, and **standard error** for each term are.
5. In the context of this data, what do these values mean?
6. Update the scatter plot you made earlier to include a fitted line.

Solutions*Question 1*

The scatterplot can be created using:

```
concrete |> ggplot(aes(x = SevenDay, y = TwentyEightDay)) + geom_point()
```

Question 2

A simple linear regression appears reasonable because the points show a clear, positive association and lie approximately along a straight-line pattern.

There is no obvious evidence of strong curvature that would make a linear model inappropriate.

Question 3

$$28\text{-Day}_i = \beta_0 + \beta_1 7\text{-Day}_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

Question 4

We can fit the model in R using

```
fit <- lm(TwentyEightDay ~ SevenDay, data = concrete )
summary(fit)
```

The resulting estimates are

Term	Estimate	Standard error
Intercept	3016.83	372.26
7-day strength	0.609	0.13

Question 5

- β_0 has no sensible physical interpretation here. 0 psi is well outside the range of the observed data, so this is an extrapolation and the intercept does not have a particularly useful practical interpretation here.
- β_1 indicates that a one pound per square inch increase in 7-Day strength produces a 0.61 pound per square inch increase in 28-Day strength.
 - Concrete gains more strength as it cures.

Question 6

```
concrete |> ggplot(aes(x = SevenDay, y = TwentyEightDay)) +
  geom_point() +
  geom_smooth(method = 'lm')
```

Simple linear regression by hand

This time, instead of using R, we will be calculating our linear regression parameters by hand. Using the concrete data we mentioned in Table 1, calculate the following:

1. $\bar{x}$, $\sum_{i=1}^n x_i^2$, and S_{xx} .
2. $\bar{y}$, $\sum_{i=1}^n y_i^2$, and S_{yy} .
3. $\sum_{i=1}^n x_i y_i$ and S_{xy} .

You may use R to verify your calculations.

However, in an exam setting you may be asked to do calculations by hand (*hint, hint, wink, wink!!*).

As such, please show all of your workings.

Use your calculated quantities to:

4. Calculate the least-squares slope estimate and the least-squares intercept estimate
5. In addition, calculate the estimated error variance.

Solutions

Questions 1, 2, and 3

$$\bar{x} = 2918.333$$

$$\sum_{i=1}^n x_i^2 = 52197900$$

$$S_{xx} = \sum x_i^2 - n\bar{x}^2 = 52197900 - 6 \times 2918.333^2 = 1097883$$

$$\bar{y} = 4795$$

$$\sum_{i=1}^n y_i^2 = 138429700$$

$$S_{yy} = \sum y_i^2 - n\bar{y}^2 = 138429700 - 6 \times 4795^2 = 477550$$

$$\sum_{i=1}^n x_i y_i = 84629400$$

$$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 84629400 - 6 \times (2918.333 \times 4795) = 668950$$

TIP: computing S_{yy} as $\sum y_i^2 - n\bar{y}^2$ is much easier than trying to compute $\sum (y_i - \bar{y})^2$. The same logic applies for S_{xx}

Question 4

$$\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{668950}{1097883} = 0.6093088 \approx 0.61$$

$$\beta_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 4795 - 0.6093088(2918.333) = 3016.834$$

Question 5

$$\sigma^2 = \frac{S_{yy} - \hat{\beta}_1 S_{xy}}{n-2} = \frac{477550 - 0.6093088 \times 668950}{6-2} = 17488.22$$